Skills Summary

Radical Equations, Domains, and Approximate Solutions

Use this reference while completing your worksheet or when studying.

Four facts everything here rests on.

- $a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$ — denominator is the root, numerator is the power.
- $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$, but $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$.
- $\sqrt{}$ returns the **non-negative** root.
- Squaring is not reversible, so it can create solutions the original equation never had.

1. Radicals and rational exponents

Read the fraction as an instruction: denominator = which root, numerator = which power. Take the **root first** to keep the numbers small.

A negative exponent means **reciprocal**, never a negative value: $a^{-p} = \dfrac{1}{a^p}$.

Example: Evaluate $25^{3/2}$ without a calculator.
Step 1. The denominator names the root and the numerator the power, so take the square root first and the arithmetic stays in two digits.
Step 2. $\sqrt{25} = 5$, and then the numerator cubes it: 5^3.

$$25^{3/2} = \mathbf{125}$$

Try it: Evaluate $64^{2/3}$ without a calculator.

check: **16**

2. Simplifying and combining radicals

Simplify by pulling out the largest **perfect-square factor**: $\sqrt{45} = \sqrt{9}\sqrt{5} = 3\sqrt{5}$.

Two radical terms combine only when the radicands match *after* simplifying — then add the coefficients and keep the radical unchanged. Radicands themselves are never added.

Example: Combine $\sqrt{18} + \sqrt{8}$ into one term.
Step 1. The radicands differ, so nothing can be decided until both radicals are in simplest form.
Step 2. $\sqrt{18} = 3\sqrt{2}$ and $\sqrt{8} = 2\sqrt{2}$, so they are like terms and the coefficients add.

$$\sqrt{18} + \sqrt{8} = \mathbf{5\sqrt{2}}$$

Try it: Combine $\sqrt{45} + \sqrt{20}$ into one term.

check: $\mathbf{5\sqrt{5}}$

3. Solving radical equations and checking

1. Isolate the radical. **2.** Note the domain (radicand ≥ 0) *and* the sign condition (the other side must be ≥ 0). **3.** Square. **4.** Solve. **5.** Substitute every candidate into the **original** equation.

Step 5 is not optional: squaring merges $A = B$ with $A = -B$, so the squared equation answers a different question than the one asked.

Example: Solve $\sqrt{x+6} = x$.
Step 1. The left side cannot be negative, so any solution needs $x \geq 0$ — note that before squaring, because squaring erases the sign.
Step 2. Squaring gives $x + 6 = x^2$, so $x^2 - x - 6 = 0$ and $(x-3)(x+2) = 0$; testing shows $x = -2$ fails the sign condition while $x = 3$ checks out.

$$x = \mathbf{3}$$

Try it: Solve $\sqrt{x+8} = x + 2$, and say which candidate is extraneous.

check: **1**

4. Approximating irrational values

Simplify **first**, approximate **last**. Rounding an intermediate radical and then multiplying spreads the error into the final digit.

Useful anchors: $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$. Sanity-check by bracketing between perfect squares.

Example: Approximate $\sqrt{11}$ to two decimal places.
Step 1. There is no perfect-square factor to remove, so bracket the value first to know what a reasonable answer looks like.
Step 2. $\sqrt{9} = 3$ and $\sqrt{16} = 4$, so the value sits just above 3; the decimal is 3.3166, which rounds to two places.

$$\sqrt{11} \approx \mathbf{3.32}$$

Try it: Approximate $2\sqrt{7}$ to two decimal places.

check: **5.29**

(!) Watch out: $\sqrt{a} + \sqrt{b}$ is never $\sqrt{a+b}$. A negative exponent reciprocates rather than negating. And a candidate that lies inside the radical's domain can still be extraneous — the sign of the *other* side is the second condition.